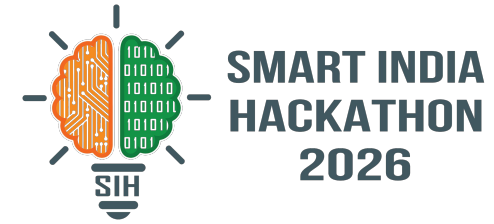
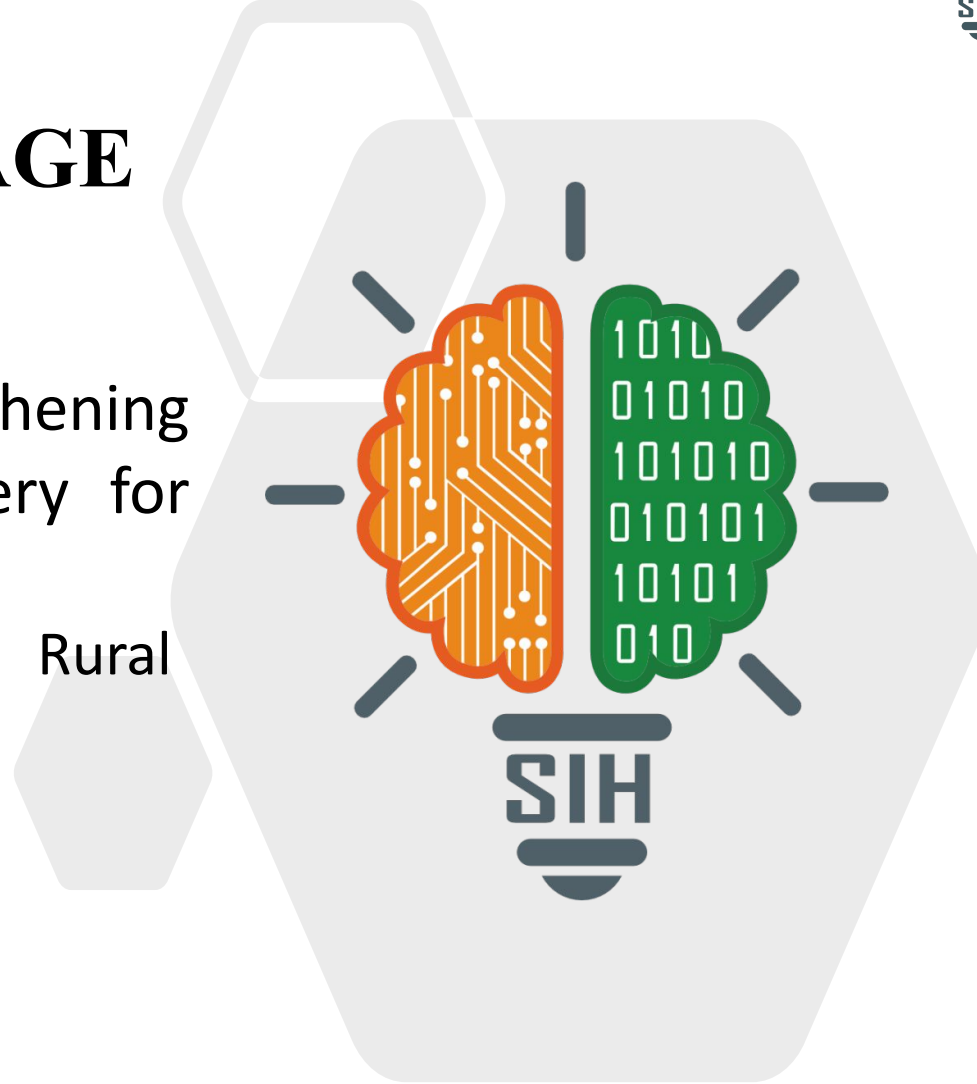


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26132
- **Problem Statement Title** - Strengthening market linkages and price discovery for farmers
- **Theme** - Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**-
- **Team Name**- MakhanChor



❖ Proposed Solution

KrishiDisha is an AI-powered market intelligence platform that converts fragmented agricultural data into **personalised selling decisions**.

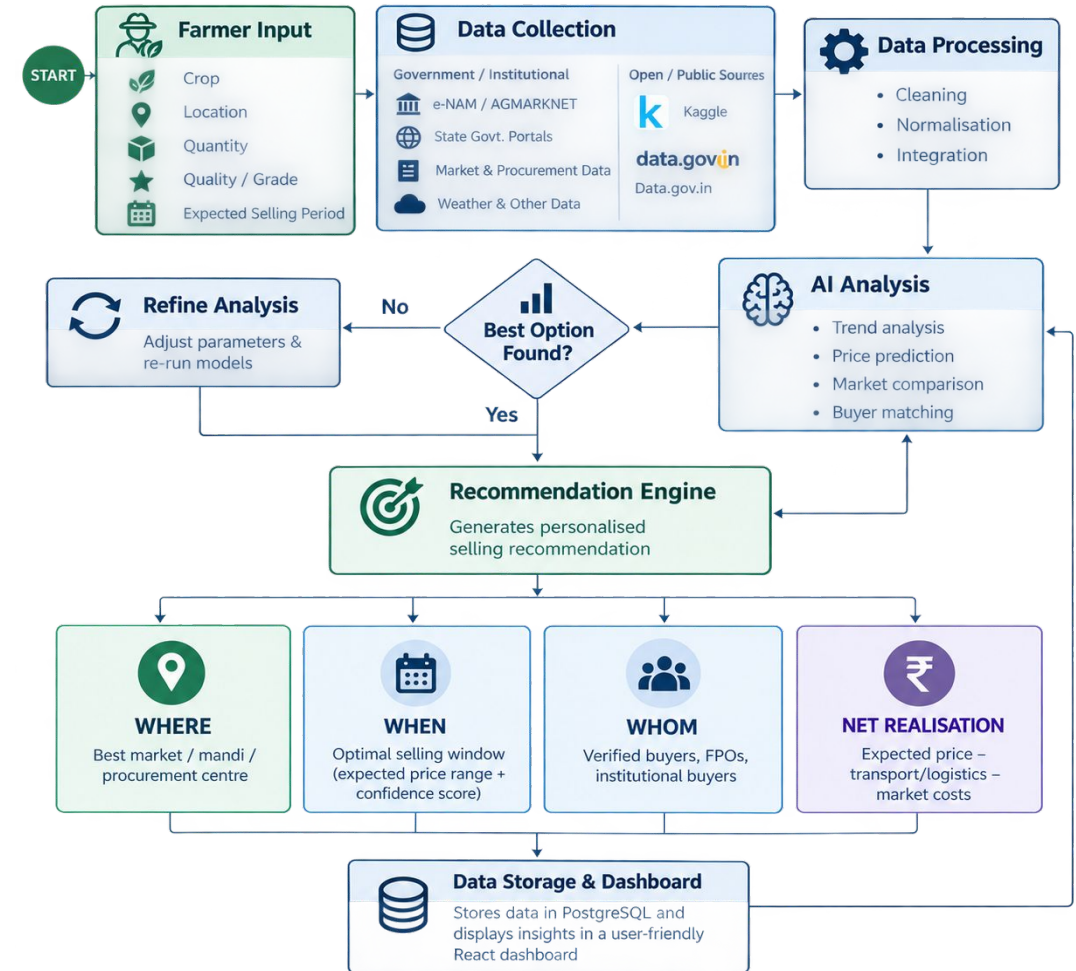
- **Market Intelligence:** Prices, arrivals, demand, buyers & historical trends
- **AI Price Analysis:** Identifies trends and estimates likely price range
- **Market & Buyer Ranking:** Compares price, distance, logistics & demand
- **Personalised Recommendation:** **WHERE • WHEN • TO WHOM** to sell

❖ How it Addresses the Problem

- **Unified Information:** Brings market, buyer & logistics data together
- **Beyond Highest Price:** Focuses on **NET REALISATION**, not just price
- **Better Timing:** Uses market trends to support selling-window decisions
- **Improved Access:** Connects farmers with **buyers, FPOs & institutions**

❖ Innovation and Uniqueness

- **Decision Support:** From raw data to actionable selling advice
- **Net Realisation Engine:** Price minus transportation/logistics cost
- **Multi-Factor Ranking:** Price + demand + distance + quality + logistics
- **Personalised AI:** Recommendations based on crop, quantity, location & timing



Tech Stack

PostgreSQL

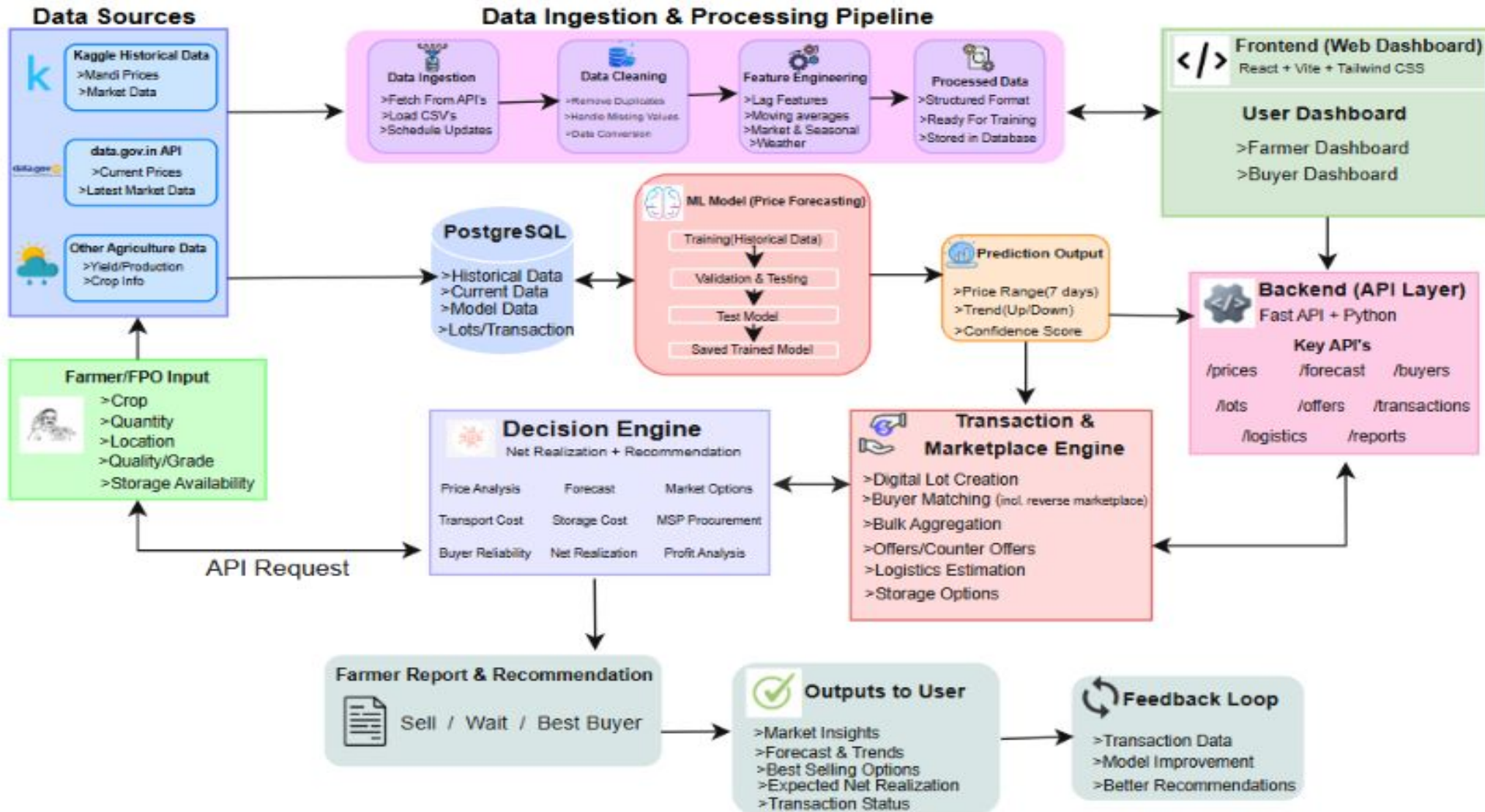
scikit
learn

FastAPI

React

tailwindcss

VITE



FEASIBILITY

Technical Feasibility

- Uses existing **e-NAM/AGMARKNET** data, geospatial services, and AI/ML for market analysis.
- Analyzes historical prices, arrivals, and seasonal trends for smart recommendations.

Operational Feasibility

- Simple user inputs: crop, quantity, location, grade, and selling timeline.
- **Multilingual**, mobile-friendly, and voice/SMS access for all digital literacy levels.

Economic Feasibility

- Leverages existing digital agri-infrastructure, minimizing setup costs.
- Optimizes market selection and transport to boost **net price realization**.

VIABILITY

Market Viability (Demand)

- Provides timely, localized market intelligence for better selling decisions.
- Serves individual farmers, FPOs, buyers, and procurement agencies.

Financial Viability (Sustainability & Revenue)

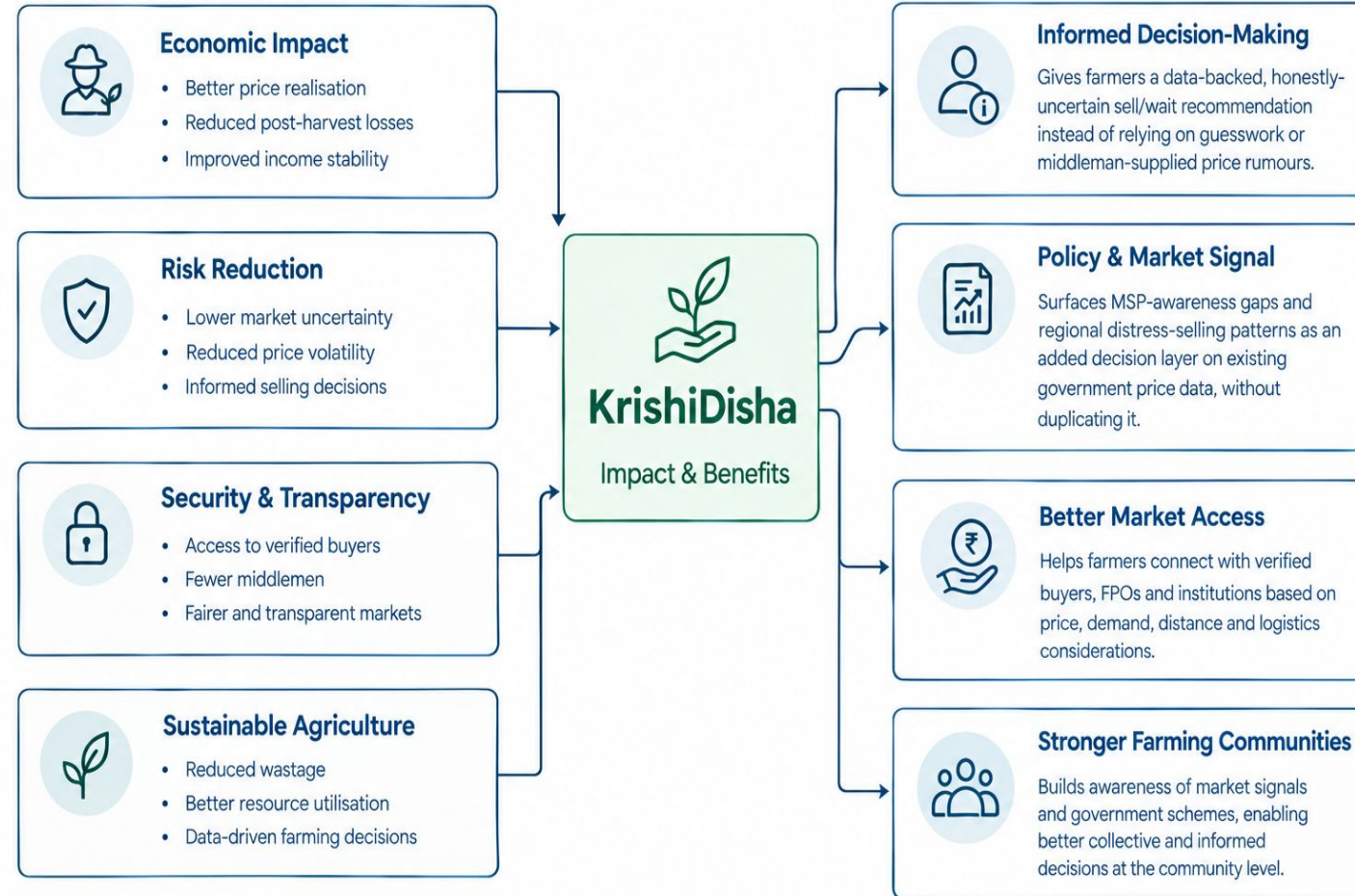
- Sustained via government, FPO, and institutional partnerships.
- Monetized through premium analytics, buyer services, and market solutions.

Social & Strategic Viability

- Enhances bargaining power and transparency by connecting verified buyers.
- Highly scalable from regional pilots across multiple crops, mandis, and states.

IMPACT AND BENEFITS

- **Higher net realization** : Knowing the prices at three nearby mandis allows a farmer to weigh the price difference against the additional fuel and transportation costs. With a calculation feature built into the platform, the farmer can identify and choose the mandi that offers the best net realisation—something that is difficult to do today without calling someone who happens to have information from that market.
- **Reduced distress selling** : Storage-aware wait guidance shifts selling away from harvest-time price dips (for crops like grains , pulses).
- **Increased trust** : Every forecast comes with a range and confidence score, not a false guarantee.
- **MSP & Procurement Awareness** : Reinforces MSP awareness and procurement uptake using existing public data.
- **Faster buyer-side sourcing** : Verified produce listings with quantity, location and quality grade cut down the time buyers/FPOs spend on manual discovery and reduce blind-buy disputes at delivery.



➤ Research Papers :

- <https://arxiv.org/pdf/1812.05173>
- <https://ideas.repec.org/a/eee/jfpoli/v81y2018icp106-121.html>
- <https://ideas.repec.org/a/eee/ejores/v311y2023i2p739-753.html>
- <https://link.springer.com/article/10.1007/s43621-025-01412-5>
- <https://www.sciencedirect.com/science/article/pii/S1877050920310449>

➤ Database Sources:

- Kaggle database link 1:
https://www.kaggle.com/code/arvindkhoda/price-of-agricultural-commodities-data/input?utm_source=chatgpt.com
- Kaggle database link 2:
<https://www.kaggle.com/datasets/thammuio/all-agriculture-related-data-sets-for-india?select=csv>

➤ Blogs and articles :

- <https://www.worldbank.org/en/news/feature/2021/03/16/a-roadmap-for-building-the-digital-future-of-food-and-agriculture>
- <https://www.fao.org/e-agriculture/es/node/14544>

➤ Deployed link :